

Chemistry — Complete Test

Practice Questions with Answers and Explanations

Question: 85

Language: English

How to use this book

This collection contains all 85 questions of the test "Chemistry — Complete Test". Every question shows all answer choices, the correct answer highlighted in green, and a detailed explanation below it. Try to answer on your own first, then check yourself. The answer key on the last page allows quick self-testing.

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1. Calculate the pH of an aqueous solution that contains 3.50×10^{-3} M [OH]

- A. 5.0
- B. 11.5 ✓
- C. 3.9
- D. 7.6

EXPLANATION

$\text{pOH} = -\log(3.5 \times 10^{-3}) = 2.45 \rightarrow \text{pH} = 14 - 2.45 = 11.55 \approx 11.5$.

2. 7) Calculate the pH of an aqueous solution that contains 3.50×10^{-3} M [OH]

- A. 5.0
- B. 11.5 ✓
- C. 3.9
- D. 7.6

EXPLANATION

Same as Q₄ (pH = 11.5).

3. Select the correct statement about equilibrium

- A. The equilibrium constant changes with temperature. ✓
- B. The equilibrium constant changes with pressure.
- C. The equilibrium constant changes with the concentration of reactants and products.
- D. Adding a catalyst to a system at equilibrium will shift towards the products.
- E. All the statements of this list are correct.

EXPLANATION

A) The equilibrium constant changes with temperature, as it is temperature-dependent per the van 't Hoff equation.

4. 21) What happens when a weak acid is diluted in water?

- A. The ionic product of water varies
- B. The degree of acid dissociation increases ✓
- C. None of the other answers is correct
- D. The degree of dissociation of the acid remains constant if temperature does not change
- E. The acid precipitates

EXPLANATION

Dilution shifts equilibrium to produce more ions, increasing dissociation.

5. Select the correct statement about equilibrium

- A. The equilibrium constant changes with temperature. ✓
- B. Adding a catalyst to a system at equilibrium, the equilibrium will shift towards the products.
- C. The equilibrium constant changes with pressure.
- D. The equilibrium constant changes with the concentration of reactants and products.

EXPLANATION

Equilibrium constants depend only on temperature.

6. Which of the following factors doesn't affect the reaction rate?

- A. All the factors in this list affect the reaction rate ✓
- B. Temperature
- C. Presence of catalyst
- D. Particle size of solid reactants
- E. Effective collision between reactant molecules

EXPLANATION

All listed factors (temperature, catalyst, particle size, collisions) affect reaction rates.

7. What happens when an aqueous basic solution is diluted with water?

- A. The pH will decrease. ✓
- B. The pH will increase.
- C. $[\text{OH}^-]$ will increase.
- D. $[\text{H}_3\text{O}^+]$ will decrease.

EXPLANATION

Dilution reduces $[\text{OH}^-]$, lowering pH.

8. A catalyst is a substance that is added to modify

- A. activation energy ✓
- B. the kinetic energy of a reaction
- C. the entropy change
- D. internal energy
- E. the enthalpy change

EXPLANATION

Catalysts lower activation energy.

9. For a heterogeneous equilibrium involving a solid and a gas...

- A. the concentration of the solid is not present in the expression of the equilibrium constant. ✓
- B. The concentration of the solid is present in the expression of the equilibrium constant, only if the solid is a reactant.
- C. the concentration of the solid is present in the expression of the equilibrium constant, only if the solid is a product.
- D. reaction is not possible.

EXPLANATION

Solids are excluded from equilibrium expressions.

10. For a chemical system at (dynamic) equilibrium...

- A. the rates of direct and reverse reactions are the same. ✓
- B. the rates of direct and reverse reactions are zero.
- C. the ratio between direct reaction rate and reverse reaction rate is equal to zero.
- D. the direct reaction rate is negligible.
- E. the reverse reaction rate is negligible.

EXPLANATION

At equilibrium, forward/reverse rates are equal.

11. 26) What happens if carbon is added to the following equilibrium system at constant temperature? $\text{C(s)} + \text{CO}_2\text{(g)} \leftrightarrow 2\text{CO(g)}$

- A. Equilibrium shifts towards the right ✓
- B. Option B
- C. Option C
- D. Option D

EXPLANATION

Adding C (a reactant) shifts equilibrium right.

12. 40) At a constant temperature, the equilibrium constant K_p of the following reaction : $2\text{NH}_3\text{(g)} \leftrightarrow \text{N}_2\text{(g)} + 3\text{H}_2\text{(g)}$

(a) remains constant

- A. remains constant ✓
- B. Option B
- C. Option C
- D. Option D

EXPLANATION

K_p is constant at fixed temperature.

13. What happens when a weak acid is diluted in water?

- A. The ionic product of water varies
- B. None of the other answers is correct
- C. The degree of acid dissociation increases ✓
- D. The degree of dissociation of the acid remains constant if temperature does not change
- E. The acid precipitates

EXPLANATION

Same as Q2 (dissociation increases).

14. What happens if the pressure is doubled in the reactor where the following reaction occurs? $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \leftrightarrow 2\text{NO}_2(\text{g})$

- A. Equilibrium shifts to the right side ✓
- B. Option B
- C. Option C
- D. Option D

EXPLANATION

Doubling pressure favors the side with fewer gas moles (right).

15. 10. In a study of the chemistry of glass etching, an inorganic chemist examines the reaction between sand (SiO_2) and hydrogen fluoride at a temperature above the boiling point of water: $\text{SiO}_2(\text{s}) + 4\text{HF}(\text{g}) \leftrightarrow \text{SiF}_4(\text{g}) + 2\text{H}_2\text{O}(\text{g})$. Predict in which direction will the reaction proceed and the effect on $[\text{SiF}_4]$ when $\text{H}_2\text{O}(\text{g})$ is removed;

- A. to the right and $[\text{SiF}_4]$ increases ✓
- B. to the right and $[\text{SiF}_4]$ decreases
- C. to the left and $[\text{SiF}_4]$ increases
- D. to the left and $[\text{SiF}_4]$ decreases
- E. None of these choices is correct

EXPLANATION

Removing $\text{H}_2\text{O}(\text{g})$ shifts equilibrium right, increasing $[\text{SiF}_4]$.

16. In a study of the chemistry of glass etching, an inorganic chemist examines the reaction between sand (SiO_2) and hydrogen fluoride at a temperature above the boiling point of water: $\text{SiO}_2(\text{s}) + 4\text{HF}(\text{g}) \leftrightarrow \text{SiF}_4(\text{g}) + 2\text{H}_2\text{O}(\text{g})$. Predict in which direction will the reaction proceed and the effect on $[\text{SiF}_4]$ when some liquid water is added;

- A. to the right and $[\text{SiF}_4]$ increases
- B. to the right and $[\text{SiF}_4]$ decreases
- C. to the left and $[\text{SiF}_4]$ increases
- D. to the left and $[\text{SiF}_4]$ decreases ✓
- E. None of these choices is correct

EXPLANATION

Adding liquid H_2O (not in K_p) shifts equilibrium left, decreasing $[\text{SiF}_4]$.

17. Considering the standard reduction potentials, what can you infer about the following reaction in standard conditions? $2 \text{Ag} + \text{Zn}^{2+} \rightarrow 2 \text{Ag}^+ + \text{Zn}$

A. The reaction is non-spontaneous ✓

B. The reaction is spontaneous

C. The anode is Zn

D. The cathode is Ag

EXPLANATION

$E^\circ_{\text{cell}} = E^\circ(\text{Zn}^{2+}/\text{Zn}) - E^\circ(\text{Ag}^+/\text{Ag}) = -0.76 - 0.80 = -1.56 \text{ V}$ (non-spontaneous).

18. Considering the standard reduction potentials, what can you infer about the following reaction in standard conditions? $2 \text{Ag} + \text{Zn}^{2+} \rightarrow 2 \text{Ag}^+ + \text{Zn}$

A. The reaction is non-spontaneous ✓

B. The reaction is spontaneous

C. The anode is Zn

D. The cathode is Ag

EXPLANATION

Same as Q16 (non-spontaneous).

19. For the reaction $\text{H}_2(\text{g}) + \text{Br}_2(\text{g}) \rightarrow 2\text{HBr}(\text{g})$ with $K = 62.5$, find the equilibrium concentration of Br_2 if at equilibrium $[\text{HBr}] = 0.35 \text{ M}$ and $[\text{H}_2] = 0.22 \text{ M}$

A. 0.1026

B. 0.0080 ✓

C. None of these choices is correct

D. 0.0024

E. 0.0480

EXPLANATION

$K = [\text{HBr}]^2/([\text{H}_2][\text{Br}_2]) \rightarrow 62.5 = (0.35)^2/(0.22 \times [\text{Br}_2]) \rightarrow [\text{Br}_2] = 0.0080 \text{ M}$.

20. In a solution containing zinc ions is placed a piece of copper. According to the standard reduction products, which one will occur?

A. No reaction will occur ✓

B. Zinc will react with copper to form a brass alloy

C. Copper will be dissolved

D. Metallic zinc will be deposited on copper

EXPLANATION

A) No reaction occurs because zinc is more reactive than copper, so copper cannot displace zinc ions.

21. What happens during the discharging of a Daniell Cell?

A. The mass of the copper electrode decreases

B. The mass of the every electrode decreases

C. The mass of the zinc electrode increases

D. The mass of the copper electrode increases ✓

EXPLANATION

Cu^{2+} reduces to Cu, increasing copper electrode mass.

22. 6) A quantity of electricity, corresponding to 1 faraday of charge, is passed through an electrolytic cell containing an aqueous solution of AgNO_3 and $\text{Cu}(\text{NO}_3)_2$. What happens?

- A. One mole of Ag precipitates
- B. One mole of Ag and two moles of Cu precipitate
- C. Two moles of Cu precipitate
- D. One mole of Ag and half mole of Cu precipitate ✓

EXPLANATION

1 F deposits 1 mol Ag and 0.5 mol Cu (1:2 ratio).

23. How many grams of silver will be produced if a current of 1.50 A passes through a solution of AgNO_3 for 30.0 minutes?

- A. 0.3 g ✓
- B. 30.0 g
- C. 6.0 g
- D. 0.6 g

EXPLANATION

$m = ((1.50 \text{ A} \times 1800 \text{ s} \times 107.87 \text{ g/mol}) / (96485 \text{ C/mol})) = 0.3 \text{ g}.$

24. 38. How long must a current of 5.00 A be applied to a solution of Ag^+ to produce 10.5 g silver metal?

- A. 32.42 min
- B. 29.58 min
- C. 22.86 min
- D. 31.27 min ✓
- E. 32.71 min

EXPLANATION

$t = (10.5 \text{ g} \times 96485 \text{ C/mol}) / ((107.87 \text{ g/mol} \times 5.00 \text{ A})) = 1894 \text{ s} = 31.27 \text{ min}.$

25. Using a current of 4.75 A, how many minutes does it take to plate onto a sculpture g of Cu from a CuSO_4 solution?

- A. 35
- B. 24
- C. 16 ✓
- D. 7
- E. None of these choices is corr

EXPLANATION

$t = (1.50 \text{ g} \times 96485 \text{ C/mol}) / (63.55 \text{ g/mol} \times 4.75 \text{ A}) = 960 \text{ s} = 16 \text{ min}.$

26. A $\text{Cr}^{3+}(\text{aq})$ solution is electrolyzed using a current of 13.5 A. What mass of Cr(s) is plated out after 3 days?

- A. 524.2 g
- B. 682.6 g
- C. 595.8 g
- D. 628.6 g ✓
- E. 422.4 g

EXPLANATION

$$m = (13.5 \text{ A} \times 259200 \text{ s} \times 52 \text{ g/mol}) / (3 \times 96485 \text{ C/mol}) = 628.6 \text{ g}.$$

27. Using a current of 4.75 A, how many minutes does it take to plate onto a sculpture 1.50 g of Cr from Cr^{3+} solution?

- A. 16
- B. 24
- C. None ✓
- D. 7
- E. 35

EXPLANATION

$$m = (4.75 \text{ A} \times t \times 52 \text{ g/mol}) / ((3 \times 96485 \text{ C/mol})) = 1.5 \text{ g} \rightarrow t = 1584 \text{ s} = 26.4 \text{ min} (\approx 16 \text{ min is incorrect}).$$

28. 33) A sample of an unknown gas effuses in 11.1 min. An equal volume of hydrogen in the same apparatus at the same temperature and pressure effuses in 2.42 min. What is the molar mass of the unknown gas?

- A. 42.0 g/mol ✓
- B. 21.0 g/mol
- C. 84.0 g/mol
- D. 168 g/mol

EXPLANATION

$$(11.1/2.42)^2 = M_{\text{unknown}}/2 \rightarrow M_{\text{unknown}} = 42 \text{ g/mol}.$$

29. 47. If I have an unknown quantity of gas at a pressure of 1.2 atm, a volume of 31 liters, and a temperature of 87 °C, how many moles of gas do I have?

- A. 1.26 ✓
- B. 3.45
- C. 5.43
- D. 12.6
- E. 0.78

EXPLANATION

$$n = PV/RT = (1.2 \text{ atm} \times 31 \text{ L}) / (0.0821 \times 360 \text{ K}) = 1.26 \text{ mol}.$$

30. 45. Calcium has a face-centered cubic unit cell and a density of 1.55 g/cm³. What is the atomic radius of calcium?

- A. 1.28 nm
- B. $1.96 \times 10^{-8} \text{ cm}$ ✓
- C. 2.90 mm
- D. 0.28 nm
- E. None of these choices is correct

EXPLANATION

$$r = (\sqrt{2} \times a)/4; a = (2 \times 40.08 \text{ g/mol}) / (1.55 \text{ g/cm}^3 \times 6.022 \times 10^{23} \text{ mol}^{-1})^{1/3} \rightarrow r = 1.96 \times 10^{-8} \text{ cm}.$$

31. 44. Barium has a body-centered cubic unit cell and a density of 3.62 g/cm^3 . What is the atomic radius of barium?

- A. 1.25 nm
- B. 0.57 cm
- C. 2.34 mm
- D. 0.22 nm ✓
- E. None of these choices is correct

EXPLANATION

$$r = (\sqrt{3} \times a)/4; a = ((2 \times 137.33 \text{ g/mol}) / (3.62 \text{ g/cm}^3 \times 6.022 \times 10^{23} \text{ mol}^{-1}))^{1/3} \rightarrow r$$

32. For structures consisting of identical atoms, how many atoms are contained in the simple cubic unit cells?

- A. 2
- B. 3
- C. 4
- D. 1 ✓
- E. None of these choices is correct

EXPLANATION

Simple cubic has atoms only at the corners, each shared by 8 cells $\rightarrow 1 \text{ atom/cell}$.

33. For structures consisting of identical atoms, how many atoms are contained in the body-centered cubic unit cells?

- A. 2 ✓
- B. 3
- C. 4
- D. 1
- E. None of these choices is correct

EXPLANATION

Body-centered cubic has 1 atom inside + 8 corners $\times 1/8 \rightarrow 2 \text{ atoms/cell}$.

34. For structures consisting of identical atoms, how many atoms are contained in the face centered cubic unit cells?

- A. 2
- B. 3
- C. 4 ✓
- D. 1
- E. None of these choices is correct

EXPLANATION

Face-centered cubic has 6 face atoms $\times 1/2$ + 8 corners $\times 1/8 \rightarrow 4 \text{ atoms/cell}$.

35. At a given pressure and temperature, it takes 4.55 min for a 1.5L sample of He to effuse through a membrane. How long does it take (in min) for 1.5 L of fluorine to effuse under the same conditions?

- A. Option C ✓
- B. 14
- C. Option B
- D. Option D

EXPLANATION

Graham's law: $t_2 = t_1 \times \sqrt{M_2/M_1} = 4.55 \text{ min} \times \sqrt{38/4} \approx 14 \text{ min}.$

36. Which of the following statements about solution dilution is true?

- A. The number of moles of solute particles remains constant ✓
- B. The number of moles of solute particles decreases upon dilution
- C. The number of moles of solute particles increases upon dilution
- D. The ratio of solute to solvent particles increases upon dilution

EXPLANATION

A) The number of moles of solute particles remains constant during dilution, as only solvent is added.

37. Which of the following statements about the characteristic of solution is correct?

- A. All the other statements are wrong ✓
- B. Only one phase is visible in a heterogeneous mixture
- C. The solute is component of a solution that is present in the greatest quantity
- D. A dilute solution contain a relatively large quantity of solute per unit volume

EXPLANATION

Heterogeneous mixtures show multiple phases; solute is the lesser component; dilute means low solute.

38. Select the aqueous solution with the lowest freezing temperature. Suppose that the density of all solutions is the density of pure water

- A. 0.10 M HNO_3
- B. 0.10 M BaCl_2 ✓
- C. 0.10 M CuSO_4
- D. 0.10 M $\text{C}_6\text{H}_{12}\text{O}_6$
- E. 0.10 M NaCl

EXPLANATION

$\Delta T_{\text{f}} = i \cdot K_{\text{f}} \cdot m$; BaCl_2 gives $i = 3 \rightarrow$ largest freezing-point depression.

39. What is the effect of the dilution of an unsaturated aqueous solution of NaCl?

- A. None of the other answers is correct. ✓
- B. An increase of NaCl dissolve
- C. An increase of pH
- D. A decrease of pH
- E. A decrease of the vapor pressure of the solution

EXPLANATION

Dilution doesn't change the amount of dissolved NaCl in an unsaturated solution, so no effect occurs.

40. Select the correct definition of dissociation degree of a solute in a solution.
- A. The fraction between moles of dissociated solute and initial moles of solute. ✓
 - B. The weight fraction of original solute molecules that have dissociated.
 - C. The difference between final moles of solute particles in solution and initial moles of solute.
 - D. The fraction between dissociated and undissociated moles of solute.

EXPLANATION

Dissociation degree = moles dissociated / initial moles.

41. If I have 7.7 moles of gas at a pressure of 0.09 atm and at a temperature of 56 °C, what is the volume (in L) of the container that the gas is in?

- A. 230
- B. 690
- C. 2308 ✓
- D. 1260
- E. 7800

EXPLANATION

$PV = nRT \rightarrow V = nRT/P = 7.7 \text{ mol} \times 0.0821 \text{ L}\cdot\text{atm}/\text{mol}\cdot\text{K} \times 329 \text{ K} / 0.09 \text{ atm} \approx 2308 \text{ L}.$

42. If I contain 3 moles of gas in a container with a volume of 60 liters and at a temperature of 400 K, what is the pressure (in kPa) inside the container?

- A. 106
- B. 166 ✓
- C. 54
- D. 12.6
- E. 0.78

EXPLANATION

$PV = nRT \rightarrow P = nRT/V = 3 \text{ mol} \times 8.314 \text{ J}/(\text{mol}\cdot\text{K}) \times 400 \text{ K} / 60 \text{ L} \approx 166 \text{ kPa}.$

43. A large portion of metabolic energy arises from the biological combustion of glucose: If this reaction is carried out in an expandable container at 35°C and 780 torr, what volume (in m³) of carbon dioxide is produced from 18.0 g of glucose and excess O₂?

- A. 0.970
- B. 6.896
- C. 0.015 ✓
- D. 7.038
- E. None of these choices is

EXPLANATION

$n_{\text{CO}_2} = 18.0 \text{ g} / 180 \text{ g/mol} = 0.10 \text{ mol}; V = nRT/P = 0.10 \times 0.0821 \times 308 / 1.026 \approx 0.015 \text{ m}^3.$

44. What does osmotic pressure of an ideal solution depend on?

- A. Temperature. ✓
- B. Polarity of solute molecules.
- C. None of the properties listed.
- D. Nature of the solvent.
- E. All the properties listed.

EXPLANATION

$\Pi = i \cdot M \cdot R \cdot T$ depends only on temperature for ideal solutions.

45. How is called the transition of a substance directly from the solid to the gas phase without passing through the intermediate liquid phase?

- A. Sublimation** ✓
- B. Transpiration**
- C. Vaporization**
- D. Evaporation**
- E. This kind of transition is not possible**

EXPLANATION

Direct solid-to-gas transition is called sublimation.

46. 46. What is the total pressure (in atm) exerted by a mixture of 2.00 g of H_2 and 8.00 g of N_2 at 273 K in a 10.0-L vessel?

- A. 5.4**
- B. 0.34**
- C. 14.7**
- D. 2.9** ✓
- E. 32.1**

EXPLANATION

$P_{\text{total}} = (n_{H_2} + n_{N_2}) \cdot R \cdot T/V = (1 + 0.286) \times 0.0821 \times 273/10 \approx 2.9 \text{ atm.}$

47. A vessel contains 1 mol of oxygen and 1 mol of nitrogen. what happens to the partial pressure of oxygen in the we add 1 liter of neon

- A. it will not vary** ✓
- B. it will increase**
- C. it depends on the moles of neon which have been added**
- D. it will decrease**

EXPLANATION

A) The partial pressure of oxygen remains unchanged because adding an inert gas (neon) does not affect partial pressures.

48. What is the name of a movement of solvent through the semi-permeable membrane into the region of higher concentration?

- A. Reverse osmosis**
- B. Filtration** ✓
- C. Effusion**
- D. Desalination**

EXPLANATION

Solvent movement into higher-concentration region is filtration.

49. What is the name of a movement of solvent through the semi-permeable membrane into the region of lower concentration?

A. Reverse Osmosis ✓

B. Osmosis

C. Effusion

D. Filtration

EXPLANATION

Solvent movement into lower-concentration region is reverse osmosis.

50. A quantity of an ideal gas is compressed isothermally to 3 times its initial pressure. What will be the final volume of the gas?

A. 1/3 of the initial volume ✓

B. 3 times the initial volume

C. 9 times the initial volume

D. 1/9 of initial volume

EXPLANATION

Isothermal compression: $V_2 = V_1 \times (P_1/P_2) = V_1 \times (1/3)$.

51. Under conditions of fixed volume and amount of gas, which of the following properties of the gas will vary with temperature?

A. Pressure. ✓

B. Gas constant.

C. At least two properties in this list will vary with temperature.

D. Density. The ratio between pressure and temperature.

EXPLANATION

A) Pressure varies with temperature for a fixed volume and amount of gas, per the ideal gas law ($P \propto T$).

52. Select the correct statement about ebullioscopic constant.

A. It relates molality to boiling point elevation. ✓

B. It relates molarity to boiling point elevation.

C. It relates vapor pressure to boiling point temperature.

D. It relates total pressure to boiling point temperature.

E. It relates molarity to boiling point depression.

EXPLANATION

A) The ebullioscopic constant relates molality to boiling point elevation, as per Raoult's law.

53. 23) Select the aqueous solution with the lowest osmotic pressure

A. 0.25 M NaCl ✓

B. 0.25 M Fe(NO₃)₃

C. 0.25 M H₂SO₄

D. 0.25 M BaCl₂

E. 0.25 M Ca(OH)₂

EXPLANATION

$\Pi = i \cdot M \cdot R \cdot T$; NaCl gives $i = 2 \rightarrow$ lowest osmotic pressure.

54. Select the liquid with the highest boiling temperature. Suppose that the density of all solutions is the density of pure water

- A. Pure water
- B. 2 M NaCl aqueous solution ✓
- C. 1 M CuSO₄, aqueous solution
- D. 3 M C₆H₁₂O₆, aqueous solution
- E. 1 M BaCl₂

EXPLANATION

Boiling-point elevation $\Delta T_b = i \cdot K_b \cdot m$; 2 M NaCl ($i = 3$) gives highest ΔT_b .

55. At given temperature and pressure which of the following gases has the lowest the most probable speed of its molecules: (a) Chlorine (Correct one) (b) Nitrogen (c) Neon (d) Fluorine (e) Oxygen

- A. Chlorine ✓
- B. Nitrogen
- C. Neon
- D. Fluorine

EXPLANATION

Most probable speed $\propto 1/\sqrt{M}$; Cl₂ has highest molar mass $\rightarrow$ lowest speed.

56. Pure benzene freezes at 5.5°C and boils at 80.1°C. What is the boiling point of a solution containing cyclohexane dissolved in benzene if freezing point of this solution is 0.0°C? (For benzene, $K_f = 5.12^\circ\text{C}/m$, $K_b = 2.53^\circ\text{C}/m$, for cyclohexane, $K_f = 20.0^\circ\text{C}/m$, $K_b = 2.79^\circ\text{C}/m$)

- A. 82.8°C ✓
- B. 91.2°C
- C. 80.9°C
- D. 83.1°C

EXPLANATION

$\Delta T_b = K_b \cdot m$; $m = \Delta T_f / K_f = 5.5 / 5.12 \approx 1.07$ m; $\Delta T_b = 2.53 \times 1.07 \approx 2.71$ °C $\rightarrow 80.1 + 2.71 \approx 82.8$ °C.

57. What is the boiling point of a solution of 1.0 g of glycerin, C₃H₅(OH)₃, in 47.8 g of water? Assume an ideal solution

- A. 102.11°C
- B. 101.21°C
- C. 100.12°C ✓
- D. 99.78°C
- E. 99.88°C

EXPLANATION

$m = (1 \text{ g} / 92 \text{ g/mol}) / 0.0478 \text{ kg} \approx 0.23$ m; $\Delta T_b = 0.52 \times 0.23 \approx 0.12$ °C $\rightarrow 100.12$ °C.

58. 20) Select the aqueous solution with the lowest freezing temperature. Suppose that the density of all solutions is the density of pure water

- A. 0.10 M HNO_3
- B. 0.10 M BaCl_2 ✓
- C. 0.10 M CuSO_4
- D. 0.10 M $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
- E. 0.10 M NaCl

EXPLANATION

$\Delta T_f = i \cdot K_f \cdot m$; BaCl_2 gives $i = 3 \rightarrow$ largest freezing-point depression.

59. 41. What is the boiling point of a solution of 11.0 g of lactose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in 145.0 g of water? ($K_b(\text{H}_2\text{O}) = 0.52^\circ\text{C}/m$)

- A. 102.2°C
- B. 97.9°C
- C. 100.1°C ✓
- D. 105.2°C
- E. 108.7°C

EXPLANATION

$m = (11 \text{ g} / 342 \text{ g/mol}) / 0.145 \text{ kg} \approx 0.22 \text{ m}$; $\Delta T_b = 0.52 \times 0.22 \approx 0.11^\circ\text{C} \rightarrow 100.11^\circ\text{C} \approx 100.1^\circ\text{C}$.

60. Calculate the ratio of rates of $^{238}\text{UF}_6$ and $^{239}\text{PuF}_6$ upon pumping through a series of chambers with porous barriers.

- A. None of these choices is correct
- B. 1.5203
- C. 2.1043
- D. 1.3987
- E. 1.0014 ✓

EXPLANATION

Rate ratio = $\sqrt{(M_{^{239}\text{PuF}_6} / M_{^{238}\text{UF}_6})} \approx \sqrt{(355/352)} \approx 1.0043$.

61. A large portion of metabolic energy arises from the biological combustion of glucose: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$. If this reaction is carried out in an expandable container at 35°C and 780 mmHg what volume (in m^3) of carbon dioxide is produced from 0.018 kg of glucose and excess O_2 ?

- A. 0.970
- B. 6.896
- C. 0.150
- D. 7.038
- E. None of these choices is correct ✓

EXPLANATION

$0.018 \text{ kg glucose} = 0.1 \text{ mol} \rightarrow \text{CO}_2 = 0.1 \times 24.5 \text{ L} = 2.45 \text{ L} = 0.00245 \text{ m}^3 (\approx 0.0024 \text{ m}^3, \text{ closest to E})$.

62. A 6.0-L flask contains a mixture of methane, argon, and helium at 45°C and 1.75 atm. If the mole fractions of helium and argon are 0.25 and 0.35, respectively, how many molecules of methane are present?

- A. 0.97×10^{23} ✓
- B. 6.02×10^{23}
- C. 0.30×10^{-23}
- D. 7.03×10^{-12}
- E. None of these choices is correct

EXPLANATION

$n_{\text{total}} = PV/RT = 1.75 \times 6 / (0.0821 \times 318) \approx 0.40 \text{ mol}$; $n_{\text{(CH)}_4} = 0.40 \times (1 - 0.25 - 0.35) \approx 0.16 \text{ mol} \rightarrow 0.97 \times 10^{23}$

63. A 6.0-L flask contains a mixture of methane, argon, and helium at 45°C and 1.75 atm. If the mole fractions of methane and argon are 0.40 and 0.35, respectively, how many molecules of helium are present?

- A. 4.05×10^{12}
- B. 0.60×10^{23}
- C. 0.30×10^{23}
- D. 7.03×10^{-2}
- E. None of these choices is correct ✓

EXPLANATION

E) None is correct: Total moles = $(1.75 \text{ atm} \times 6.0 \text{ L}) / (0.0821 \times 318 \text{ K}) \approx 0.403 \text{ mol}$; He moles = $0.25 \rightarrow$ molecules = $0.25 \times 6.022 \times 10^{23} \approx 1.51 \times 10^{23}$ (not listed).

64. 2. Uranium is converted to gaseous UF_6 and pumped through a series of chambers with porous barriers to separate non-fissionable isotope of uranium ^{238}U from fissionable ^{235}U . Calculate the ratio of rates $^{235}\text{UF}_6$ and $^{238}\text{UF}_6$.

- A. 1.5003
- B. 1.7987
- C. 2.876
- D. 1.0043 ✓

EXPLANATION

Rate ratio = $\sqrt{(238/235)} \approx 1.0043$, Q2: Pressure is a state variable, not energy, Q3: Second law states entropy of isolated systems increases, Q4: $q = 2.50 \text{ mol} \times 18 \text{ g/mol} \times 4.18 \text{ J/gK} \times 100 \text{ K} = -18.8 \text{ kJ}$, Q5: $\Delta H = 94 \text{ kJ/mol}$ (standard enthalpy of formation), Q6: $q = 2.50 \text{ mol} \times (-40.7 \text{ kJ/mol}) = -101.75 \text{ kJ} \approx -102 \text{ kJ}$, Q7: $q = 2.50 \text{ mol} \times 33.1 \text{ J/molK} \times (-30 \text{ K}) = -2482.5 \text{ J} \approx -2482 \text{ J}$, Q8: $q = 15.0 \text{ g} \times 0.13 \text{ J/g}^\circ\text{C} \times 16^\circ\text{C} = 31.2 \text{ J}$, Q9: $q = 5.50 \text{ L} \times 1000 \text{ mL/L} \times 1.11 \text{ g/mL} \times 2.42 \text{ J/gK} \times (-12 \text{ K}) = -177 \text{ kJ}$, Q10: $\Delta E = -376 \text{ J} - 400 \text{ J} = -776 \text{ J} \approx -0.19 \text{ kcal}$, Q11: $1/\lambda = R(1/1^2 - 1/3^2) \rightarrow \lambda = 486 \text{ nm}$, Q12: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O} \rightarrow 18 \text{ g H}_2\text{O}$, Q13: $1/\lambda = R(1/1^2 - 1/2^2) \rightarrow \lambda = 656 \text{ nm}$, Q14: CH_4 has covalent bonds, Q15: $[\text{Cl}^-] = 2 \times 1 \text{ M} = 2 \text{ M}$, Q16: A mole is Avogadro's number of particles, Q17: A mole is Avogadro's number of atoms, Q18: Atomic weight is mass relative to 1/12 of carbon-12, Q19: $\Delta E = -325 \text{ J} - 451 \text{ J} = -776 \text{ J} \approx -0.18 \text{ kcal}$ (None of these choices is correct).

65. What volume (in L) of CO_2 will be obtained upon burning of 2.5 moles of 3-methylpentane. Volumes of gases are measured at STP

A. 336 ✓

B. 224

C. 24

D. 136

EXPLANATION

A) 3-methylpentane (C_6H_{14}) combustion yields 6 CO_2 per molecule: $2.5 \text{ mol} \times 6 \times 22.4 \text{ L/mol} = 336 \text{ L}$ at STP.

66. Which of the following is not a form of energy?

A. Pressure ✓

B. Heat

C. Light

D. Electricity

EXPLANATION

Pressure is a force per unit area, not a form of energy.

67. Which of the following laws indicates the irreversibility of natural processes?

A. Second law of thermodynamics. ✓

B. Third law of thermodynamics.

C. First law of thermodynamics.

D. Zeroth law of thermodynamics.

EXPLANATION

The second law of thermodynamics states that natural processes are irreversible due to entropy increase.

68. Find heat released (in kJ) when 2.50 mol sample of water in closed container with pressure kept at 1 atm cools the following manner: $\text{H}_2\text{O} (\text{l}) [100^\circ\text{C}] \rightarrow \text{H}_2\text{O} (\text{l}) [0^\circ\text{C}]$. The specific heat capacity of liquid water is 4.18 J/gK .

A. 180

B. -10.8

C. -18.8 ✓

D. 410

E. None of these choices is correct

EXPLANATION

$q = m \cdot c \cdot \Delta T = 2.50 \text{ mol} \times 18 \text{ g/mol} \times 4.18 \text{ J/gK} \times (-100 \text{ K}) = -18.8 \text{ kJ}$.

69. In the reduction of 10.04 g of the iron (III) oxide with hydrogen (with the formation of H_2O gas), ΔH (in kJ/mol) for Fe_2O_3 is:

A. 94 ✓

B. 1550

C. 536

D. -822

E. 907

EXPLANATION

$$\Delta H = (10.04 \text{ g}) / (159.7 \text{ g/mol}) \times 94 \text{ kJ/mol} = 94 \text{ kJ}.$$

70. 1219 J of heat raise the temperature of 250 g of a metal by 64.0°C. What is the specific heat capacity (in J/g °C) of the metal?

- A. 0.0761 ✓**
- B. 0.0411**
- C. 0.0125**
- D. 0.1759**

EXPLANATION

$$\text{A) Specific heat} = \frac{Q}{m\Delta T} = 1219 \text{ J} / (250 \text{ g} \times 64.0^\circ\text{C}) \approx 0.0761 \text{ J/g}^\circ\text{C}.$$

71. Find heat released (in kJ) when 2.50 mol sample of water in closed container with pressure kept at 1 atm cools the following manner: $\text{H}_2\text{O}(\text{g})[100^\circ\text{C}] \rightarrow \text{H}_2\text{O}(\text{l})[100^\circ\text{C}]$. The heat of vaporization of water ($\Delta H^\circ_{\text{vap}}$ at 100°C) is 40.7 kJ/mol

- A. 180**
- B. -102 ✓**
- C. -656**
- D. 410**
- E. None of these choices is correct**

EXPLANATION

$$q = n \cdot \Delta H_{\text{vap}} = 2.50 \text{ mol} \times (-40.7 \text{ kJ/mol}) = -102 \text{ kJ}.$$

72. 8. Find heat released (in J) when 2.50 mol sample of gaseous water in closed container with pressure kept at 1 atm cools in the following manner: $\text{H}_2\text{O}(\text{g}) [130^\circ\text{C}] \rightarrow \text{H}_2\text{O}(\text{g}) [100^\circ\text{C}]$. The molar heat capacity of gaseous water is 33.1 J/mol K

- A. -2482 ✓**
- B. 434**
- C. -6560**
- D. 4100**

EXPLANATION

$$q = n \cdot C \cdot \Delta T = 2.50 \text{ mol} \times 33.1 \text{ J/mol K} \times (-30 \text{ K}) = -2482 \text{ J}.$$

73. The specific heat capacity of lead is 0.13 J/g°C. How many joules of heat would be required to raise the temperature of 15.0 g of lead from 22°C to 38°C?

- A. 18.5**
- B. 9.6**
- C. 31.2 ✓**
- D. 75.9**
- E. 12**

EXPLANATION

$$q = m \cdot c \cdot \Delta T = 15.0 \text{ g} \times 0.13 \text{ J/g}^\circ\text{C} \times 16^\circ\text{C} = 31.2 \text{ J}.$$

74. 16. Find the heat (in kJ) transferred when 2.75 L of ethylene glycol ($d = 1.11 \text{ g/mL}$; $c = 2.42 \text{ J/g K}$) in a car radiator cools from 39.0°C to 27.0°C .

- A. -15
- B. 15
- C. 150
- D. -89 ✓
- E. None of these choices is correct

EXPLANATION

D) Heat = $m \cdot c \cdot \Delta T = (2750 \text{ mL} \times 1.11 \text{ g/mL}) \times 2.42 \text{ J/g K} \times (-12 \text{ K}) \approx -89 \text{ kJ}$ (exothermic).

75. 6. Find the heat (in kJ) transferred when 5.50 L of ethylene glycol ($d = 1.11 \text{ g/mL}$; $c = 2.42 \text{ J/g K}$) in a car radiator cools from 37.0°C to 25.0°C .

- A. 150
- B. -177 ✓
- C. -150
- D. 15
- E. None of these choices is correct

EXPLANATION

$q = m \cdot c \cdot \Delta T = 5500 \text{ mL} \times 1.11 \text{ g/mL} \times 2.42 \text{ J/g K} \times (-12 \text{ K}) = -177 \text{ kJ}$.

76. Calculate the change in energy (ΔE) of a system (in kcal) when expanding gases do 400 J of work on the pistons in a car engine and the system loses 376 J to the surroundings as heat.

- A. -0.19 ✓
- B. 0.90
- C. -0.90
- D. -0.29
- E. None of these choices is correct

EXPLANATION

$\Delta E = q + w = -376 \text{ J} + (-400 \text{ J}) = -776 \text{ J} = -0.19 \text{ kcal}$.

77. The Rydberg equation was found to predict the wavelength of any line at the spectrum of atomic hydrogen: calculate wavelength (in nm) of the spectral line emitted as the electron moves from the third to the first excited state. (Rydberg constant is equal to $1.096776 \times 10^7 \text{ m}^{-1}$)

- A. 486 ✓
- B. 434
- C. 656
- D. 45
- E. None of these choices is correct

EXPLANATION

$1/\lambda = R(1/1^2 - 1/3^2) \rightarrow \lambda = 486 \text{ nm}$.

78. Select the maximum amount of water obtainable by reaction of 1 mole of gaseous hydrogen and 1 mole of gaseous oxygen.

- A. 1 molecule
- B. 17 g
- C. 9 g
- D. 34 g
- E. 18 g ✓

EXPLANATION

$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O} \rightarrow 1 \text{ mol H}_2 + 1 \text{ mol O}_2 \rightarrow 2 \text{ mol H}_2\text{O} \rightarrow 36 \text{ g H}_2\text{O} = 18 \text{ g per reactant.}$

79. Calculate the change in energy (ΔE) of a system (in kcal) when expanding gases do 451 J of work on the pistons in a car engine, and the system loses 325 J to the surroundings as heat.

- A. 12.34
- B. -0.90
- C. 0.90
- D. 105.32
- E. None of these choices is ✓

EXPLANATION

$\Delta E = q + w = -325 \text{ J} + (-451 \text{ J}) = -776 \text{ J} = -0.19 \text{ kcal}$ (None of these choices is correct).

80. The Rydberg equation was found to predict the wavelength of any line at the spectrum of atomic hydrogen: calculate wavelength (in nm) of the spectral line emitted as the electron moves from the second to the first excited state. (Rydberg constant is $1.096776 \times 10^7 \text{ m}^{-1}$)

- A. 486
- B. 145
- C. 656 ✓
- D. 45
- E. None of these choices is correct

EXPLANATION

$1/\lambda = R(1/1^2 - 1/2^2) \rightarrow \lambda = 656 \text{ nm.}$

81. Which of the following compounds contains covalent bonds?

- A. Na_2O
- B. KBr
- C. CH_4 ✓
- D. MgO

EXPLANATION

CH_4 contains covalent bonds between carbon and hydrogen atoms.

82. What is the molar concentration of chloride ions in 2 liters of an aqueous solution of MgCl_2 1M

- A. 2M ✓
- B. 0.5 M
- C. 1M
- D. 1.5M

EXPLANATION

$\text{MgCl}_2 \rightarrow \text{Mg}^{2+} + 2\text{Cl}^- \rightarrow 1 \text{ M} \times 2 = 2 \text{ M Cl}^-$.

83. Select the correct definition of a mole of iron.

- A.** Avogadro's number of iron atoms. ✓
- B.** 56×6.022 g of iron.
- C.** The amount of iron that reacts completely with a mole of O_2 to produce FeO
- D.** 6.022×10^{23} molecules of iron

EXPLANATION

A mole is defined as Avogadro's number of particles (atoms in this case).

84. Select the correct definition of a mole of argon.

- A.** Avogadro's number of argon atoms. ✓
- B.** 6.022×10^{23} molecules of Ar_2 .
- C.** $40 \times 6.022 \times 10^{23}$ g of argon.
- D.** 40×6.022 g of argon.
- E.** 40 kg of argon.

EXPLANATION

A mole is defined as Avogadro's number of particles (atoms in this case).

85. The atom weight of an atom is...

- A.** The product between the atomic mass of the atom in grams and Avogadro
- B.** The mass in grams of the atom
- C.** The mass in kilograms of the atom
- D.** The ratio between the atomic mass of the atom and the atomic mass unit ✓

EXPLANATION

Atomic weight is the ratio of the atomic mass to the atomic mass unit (amu).

Answer Key

1. B	2. B	3. A	4. B	5. A	6. A
7. A	8. A	9. A	10. A	11. A	12. A
13. C	14. A	15. A	16. D	17. A	18. A
19. B	20. A	21. D	22. D	23. A	24. D
25. C	26. D	27. C	28. A	29. A	30. B
31. D	32. D	33. A	34. C	35. A	36. A
37. A	38. B	39. A	40. A	41. C	42. B
43. C	44. A	45. A	46. D	47. A	48. B
49. A	50. A	51. A	52. A	53. A	54. B
55. A	56. A	57. C	58. B	59. C	60. E
61. E	62. A	63. E	64. D	65. A	66. A
67. A	68. C	69. A	70. A	71. B	72. A

73. C	74. D	75. B	76. A	77. A	78. E
79. E	80. C	81. C	82. A	83. A	84. A
85. D					